Discover Applied Sciences

Research

Oxygen-modulated photoresponse in nickel oxide thin films for wide band gap photodetector application

Swati Suman[1] · **Anupama Behra**[1] · **Parasuraman Swaminathan**[1,2]

Received: 16 September 2024 / Accepted: 7 January 2025
Published online: 15 January 2025
© The Author(s) 2025 OPEN

Abstract

Metal oxides, due to their wide band gap, are well suited for use in photodetectors as the active light-absorbing layer. While there have been extensive studies on *n*-type oxides, such as tungsten oxide and zinc oxide, there is relatively little work on the use of *p*-type oxides, such as nickel oxide (NiO), for photodetectors. Using these *p*-type oxides along with *n*-type conducting oxides to form heterojunctions can improve the detection capabilities by efficient separation of charge carriers. In this work, the photoresponse of a *p*-type NiO thin film, grown by room temperature reactive magnetron sputtering from a pure nickel target onto a conducting *n*-type indium-doped tin oxide (ITO) substrate, is investigated. Different ratios of oxygen to argon (10/45, 15/45, and 20/45) during sputtering are investigated, and the effect of oxygen concentration on the optical properties of the NiO film is studied, which helps in modulating the photoresponse of the developed detector. The O_2/Ar ratio, with a value of 15/45, is found to perform better among the three ratios. For the corresponding photodetector, the current under illumination, is found to be nearly three orders of magnitude higher than the dark current, even at a very low applied voltage of 0.1 V. The calculated responsivity (at 0.012 A/W) is found to be the best among all three values. The device also has an excellent transient current response with a rise time of 0.5 s and a decay time of 1.4 s, which is the fastest transient behavior among all three ratio-based devices. The work paves the way for using metal oxide-based heterojunctions as wide band gap photodetectors.

Highlights

- Nickel oxide films grown by reactive magnetron sputtering
- A wide band gap heterostructure-based photodetector is fabricated using nickel oxide
- Effect of oxygen gas on the photodetector properties is evaluated

Keywords Nickel oxide · Magnetron sputtering · Photoresponse · Oxygen concentration · Heterojunction

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s42452-025-06485-5.

✉ Parasuraman Swaminathan, swamnthn@iitm.ac.in | [1]Electronic Materials and Thin Films Lab, Department of Metallurgical and Materials Engineering, Indian Institute of Technology (IIT) Madras, Chennai, India. [2]Centre of Excellence in Ceramics Technologies for Futuristic Mobility, Indian Institute of Technology (IIT) Madras, Chennai, India.

1 Introduction

Wide band gap materials, such as metal oxides, have a significant role in developing optoelectronic devices, which includes wide band gap photodetectors, flexible electronic devices [1], and solar cells, etc. [2–5] These metal oxide band gaps (typically greater than 3 eV) can be tuned with doping and other structural modifications [6], creating strong electrical conductivity and high electron mobility [7], making them suitable for wide-bandgap photodetectors. These photodetectors have the capability of absorbing radiation over a wavelength range that includes the ultraviolet (UV) and the visible spectrum with applications such as biological and chemical sensors, flame and combustion engine monitoring, military missile detection, and UV radiation detection, to name a few [8–12]. Their advantageous properties include low permittivity, high avalanche breakdown, good thermal conduction, high electron saturation rates, and resistance to high energy radiation. These properties make these materials suitable for developing robust electronic devices that can work at high temperatures and at high powers and frequencies [13–15].

Metal oxides can be broadly classified as *n*- and *p*-type semiconductors, and both have been studied by researchers for wide-bandgap photodetector applications. Some common examples include oxides such as TiO_2, ZnO, SnO_2, V_2O_5, NiO, and Nb_2O_5 [16–24]. Among the various growth techniques available, recently, a variety of low-cost growth techniques, such as sol–gel have also been developed for synthesizing these oxides [23, 25–27]. Metal oxides provide good responsivity to UV irradiance [17] and good stability at room and elevated temperatures [28]. NiO (a *p*-type semiconductor) with a band gap range of 3.5–4.0 eV [29] is non-toxic, earth-abundant, transparent to visible light, and has good thermal and chemical stability [12]. It has been studied extensively for its electronic and optical properties [30]. Considering these promising aspects, NiO has been investigated towards developing a wide band gap photodetector [12], photovoltaics [31], resistive switches [32], and resistive random access memory devices [33].

Out of a wide range of thin film deposition techniques, the reactive DC sputtering technique has tremendous benefits over other vacuum deposition techniques. It gives better and more precise control of film by tunning other parameters like power, gas flow rate, different gases and combinations of gases that can be used, temperature control, deposition time, and distance control between the target and substrate, etc. [35, 36]. The versatile nature of the reactive DC sputtering technique can be employed to develop uniform coverage, better control over the intended compound stoichiometry, and large-area application. It is also suitable for many applications, from microelectronic fabrication to decorative coatings.

The oxygen concentration in NiO plays an important role in determining its electrical and optical properties [34, 35]. Primarily, in reactive sputtering, oxygen concentration can influence the morphology, thickness, and electrical and optical properties of the film [39, 40]. Sun et al. studied the effect of sputtering NiO under pure oxygen flow, varying the working pressure from 0.1 to 2 Pa and showed that increasing oxygen content helped grow the films with better crystallinity having fewer trap centres. However, the sputtering power used was high (600 W) making the whole growth process power intensive [41]. Kim et al. have investigated the influence of oxygen flow rate from 0–10 sccm (keeping the Ar flow of 20 sccm) with a base/working pressure of $5 \times 10^{-6}/5 \times 10^{-3}$ Torr. They were able to demonstrate that increasing oxygen contributes to better hole transport and, hence, carrier mobility [36]. Similar studies have been carried out for other reactive sputtered films such as nitrides [37], borides [38], carbides [39], and chalcogenides films [40, 41].

There have been studies on the importance of heterostructure for enhancing the performance of photodetectors, in both photoconductor and vertical geometry. There are several combinations of materials used for forming heterostructures, such as ZnO/TiO_2 [25], SnO_2/TiO_2 [26], ITO/SnS [42] in photoconductor geometry and n-VO_2/n-$MoSe_2$[43], p-NiO/n-ZnO [44], TiO_2/p-Si [45] in vertical geometry. The heterojunction also brings properties such as self-bias [41, 42] and self-rectifying resistive switching [32] to the device, which has been demonstrated earlier. Moreover, the photoresponse can further be enhanced by including plasmon resonance [46].

In this article, we report on the performance of a NiO-based wide band gap photodetector and the effect of the oxygen pressure, during growth, on the optoelectronic properties of the NiO film. A photoconductor geometry, by depositing *p*-type NiO over an *n*-type ITO-coated substrate to form a heterostructure, was developed, and the response was measured between two parallel silver electrodes deposited on top of NiO. Higher oxygen concentration helps in reducing sheet resistance and increases conductivity; this gives an opportunity to select oxygen flow from 10, 15, and 20 sccm as most investigations has been carried out for lower oxygen flow rates (less than 10 sccm). Photoresponse of the device was examined at different bias voltages, and the device demonstrated a large change in current under illumination, even at a very low bias voltage of 0.1 V. Besides, transient photocurrent was studied

to examine the rate of photo-responsiveness of the device. The results have been explained based on the role of oxygen on the photoresponse properties of the photodetector.

2 Experimental section

2.1 Material synthesis and device fabrication

In this study, ITO-coated glass substrates (1.5 cm × 1.5 cm) with sheet resistance of approximately 20 Ohm/sq. and with transmittance up to 85% at 550 nm were purchased from Technistro, India. These were cleaned thoroughly with soap solution and rinsed by DI water ultrasonication for 15 min, followed by ultrasonication cleaning in acetone and then isopropanol (IPA) for 15 min each. UV-Ozone treatment was carried out after drying the substrate with nitrogen gas to create a hydrophilic surface. Then, the samples were placed in a reactive DC magnetron chamber (HHV ATS-500 supplied by Hind High Vacuum) for film growth. The distance between the substrate and the nickel target was maintained at 10 cm during the deposition process. The substrates were degassed for 30 min to eliminate impurities. Then, pre-sputtering was done at 150 W to clean the target. Here, nickel metal (purity 99.95%) (Ni/V: (93/7 wt.%)) with a diameter of 5 cm and thickness of 6 mm, purchased from Beijing Goodwill, was used as the target to develop a nickel oxide film in the presence of argon (carrier gas) and oxygen (reactive gas) gases. The flow rate of argon was fixed at 45 standard cubic centimetres per minute (sccm), while the oxygen flow rate was varied with values of 10, 15, and 20 sccm to create three different types of NiO films. The sputtering power was kept constant at 150 W, and the sputtering time was kept constant at 30 min for all the films. Sputtering was carried out with the substrate kept at room temperature (25 °C). The chamber base pressure was kept 7×10^{-3} mbar while the working pressure was maintained at 9×10^{-3} mbar. The sputtered NiO film was subsequently annealed at 450 °C, in ambient conditions, for 60 min to generate a pinhole-free thin film. To fabricate the device, parallel electrical contacts were developed on top of the NiO film-coated ITO substrate using silver epoxy. The epoxy was dried at 65 °C for 10 min. The device fabrication steps are schematically shown in Fig.1. For testing the opto-electronic properties, the light was illuminated on the device from the top side, and the generated charge carriers were collected at different applied biases.

2.2 Characterization techniques

X-ray diffraction (XRD) analysis of thin films was carried out using a Rigaku SmartLab X-Ray diffractometer (Cu Kα radiation with wavelength of 0.154 nm), and high-resolution scanning electron microscopy (HRSEM) was performed using a Field-Emission SEM (FE-SEM) Quanta 3D. UV-Vis absorption spectroscopy (200–900 nm) was carried out using a Jasco V-650 spectrophotometer. Raman spectroscopy (using radiation of wavelength 632 nm) was done using a BRUKER RFS 27 MultiRAM FT Spectrometer. Surface roughness was studied by using atomic force microscopy (AFM) (BRUKER-DIMENSION icon with ScanAsyst). Current–voltage (I–V) curves and photoelectric responses of the photodetectors were measured using a source meter (Keysight B2912A) with a solar simulator (Sciencetech SLB-300-A Compact Xe arc lamp) source of wavelength range from 100–700 nm with 1000 W/m^2 intensity (AM 1.5 G) with a coverage diameter of 70 mm.

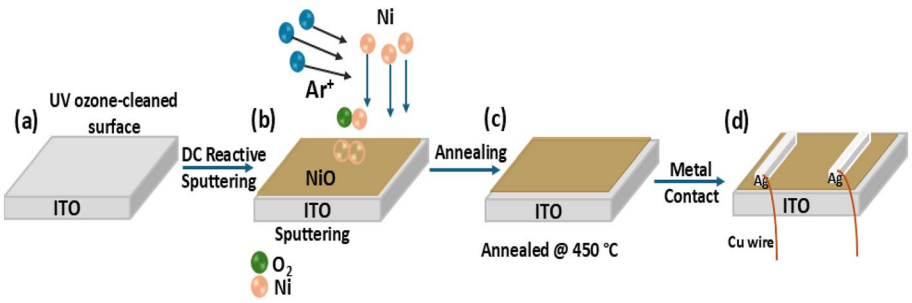


Fig. 1 Schematic of the device **a** cleaned ITO substrate, **b** NiO thin film deposition on ITO substrate using reactive sputtering, **c** annealing of the grown NiO film, and **d** metal contact development on top of the film. The oxygen flow rate was varied in step **b** to produce different types of NiO, while the rest of the steps remained the same

3 Results and discussion

3.1 Structural and morphological characterization

The structure of the sputtered NiO film on ITO was investigated by XRD, for three different ratios of argon to oxygen, and Fig. 2a summarises the plots. Sharp diffraction peaks are evident, which can be indexed to NiO in a face-centred cubic structure (ICDD no. 98-055-9068). Specifically, the two peaks found roughly at 36.8° and 43.9° can be indexed to the (111) and (200) planes of NiO [47, 48]. These peaks have been magnified and shown in the inset of Fig. 2(a). The remaining peaks correspond to ITO from the substrate (ICDD no. 01-089-4597). To distinguish more clearly the NiO peaks from ITO, bare NiO and ITO were analyzed by XRD and the XRD plots are shown in Figure S1 in Supporting Information file. The cross-sectional SEM images of the films are shown in Figures 2b–d, with the insets providing the top view. The images show that as the concentration of oxygen is increased, the thickness of the sputtered film increases. The measured thickness from the cross-sectional images shows values of 195 nm (10 sccm of O_2), 218 nm (15 sccm of O_2), and 284 nm (20 sccm of O_2) for the different amounts of oxygen. This increase in thickness with oxygen flow rate can be attributed to the fact that at higher oxygen partial pressure, a greater number of sputtered Ni species (which depends on the flow rate of Ar) will react with oxygen to form NiO. The inset images show that the surface of the film is uniform and crack-free post-annealing at 450 °C for 60 min. To investigate further the annealing effect on film morphology, low-magnification SEM images of the sample surface have been recorded prior to annealing, which can be seen in Fig.S2. The cracks are more prominent and can be seen clearly, which has been improved post-annealing. Annealing promotes grain growth by increasing particle size at higher temperatures and this leads to a reduction in crack density.

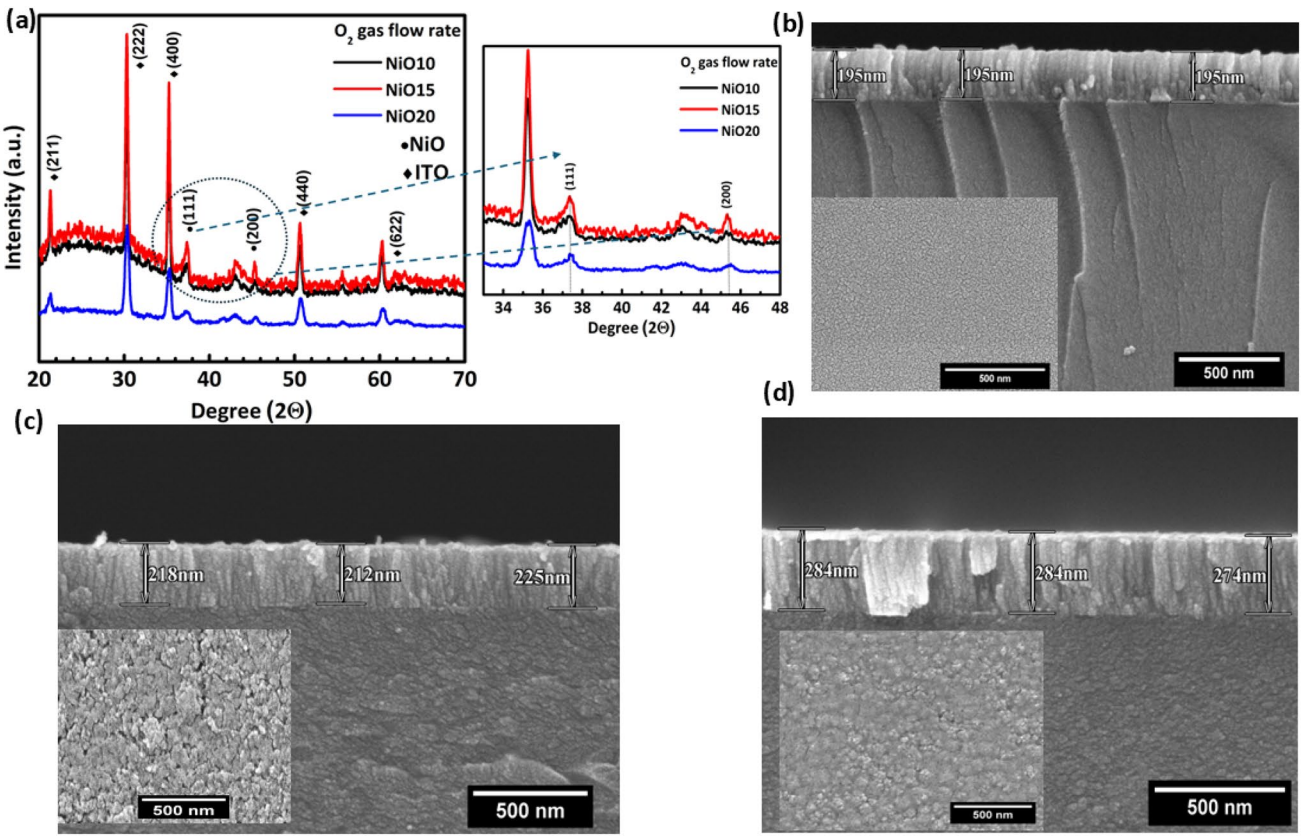


Fig. 2 **a** XRD peaks of NiO films sputtered on ITO substrate at three different oxygen flow rates. Peaks corresponding to both NiO and ITO have been shown, and the inset shows the magnified 2theta region for the (111) and (200) peak positions. The plots for bare NiO and ITO are given in Fig. S1. **b** Cross-sectional images of the film showing the thickness of the NiO film on ITO substrate with **b** 10, **c** 15, and **d** 20 sccm flow rate of O_2. The inset shows the top view SEM images of NiO on ITO substrates for all three oxygen flow rates

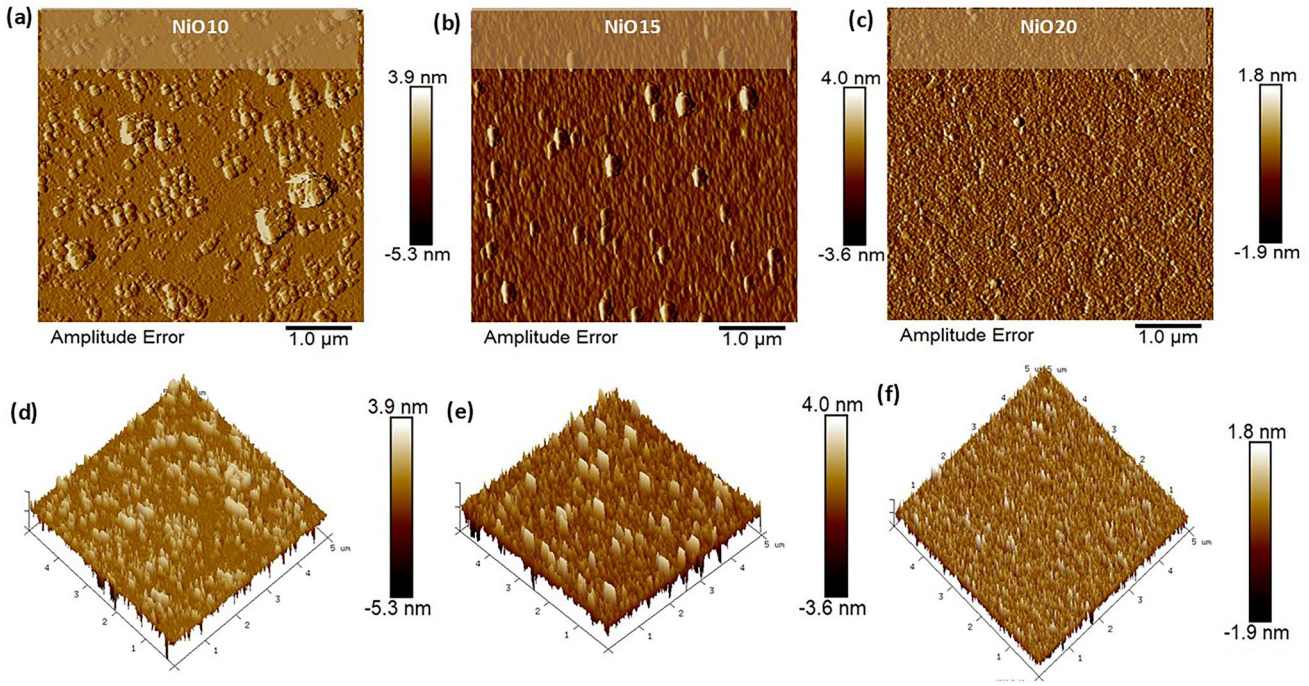


Fig. 3. 2D and 3D AFM micrographs of the NiO film deposited on ITO substrate with oxygen flow rates of **a**, **d** 10, **b**, **e** 15, and **c**, **f** 20 sccm. With increasing O_2 concentration, the films become smoother

The AFM images in Fig. 3 show an irregular shape and randomly oriented grains captured over an area of 5×5 µm^2, and the surface roughness was determined through image analysis using Nanoscope AFM software. As the amount of oxygen increased, there was a slight decrease in the roughness, from an average roughness size of 3.73 nm (10 sccm O_2) to 3.75 nm (15 sccm O_2), and 3.36 nm (20 sccm O_2). From the AFM images, it can be seen that as the concentration of oxygen increases, the films become smoother.

3.2 Optical characterization

For optical studies, the sputtered-coated NiO on ITO was investigated by UV-Vis spectroscopy. The observed spectra for all three compositions are shown in Fig. 4a, with absorbance maxima located in the UV region. The optical band gaps of the grown film were calculated using the Tauc plot and equation as given below in equations 1 and 2 [49].

$$(\alpha h\nu)^2 = A(h\upsilon - Eg) \tag{1}$$

$$\alpha = \left(1/t\right) ln \frac{1}{T} \tag{2}$$

where α is the absorption coefficient, *h* is plank constant, υ is the frequency of the incident radiation, *t* is the film thickness, and *A* is a constant [30]. As the film thickness increases, the absorption coefficient decreases. The Tauc plots are given in Fig. 4b and the corresponding optical gap values are found to be 3.52, 3.56, and 3.68 eV, with increasing amounts of oxygen. This increase in the optical band gap of the NiO films with increasing oxygen concentration can be attributed to the decrease in shallow defects at greater oxygen flow rates. The transparency of the coated films can be seen in Fig. 4c, while the corresponding transmittance plots are combined in Fig. 4d, indicating that the transmittance is reduced as the O_2 concentration increases. This is directly related to the increase in the thickness of the NiO film at higher oxygen concentrations, and can be seen in the cross-sectional SEM image in Fig. 2b–d.

The Raman spectra for the three samples are compared in Fig. 4e. The first-order longitudinal (LO) mode is located at 562.4 cm^{-1}, while the stronger second-order band represented by a transverse mode (2TO) is located at 758.1 cm^{-1}. The band around 895 cm^{-1} corresponds to the combination of TO + LO, while another longitudinal (2LO) mode was found

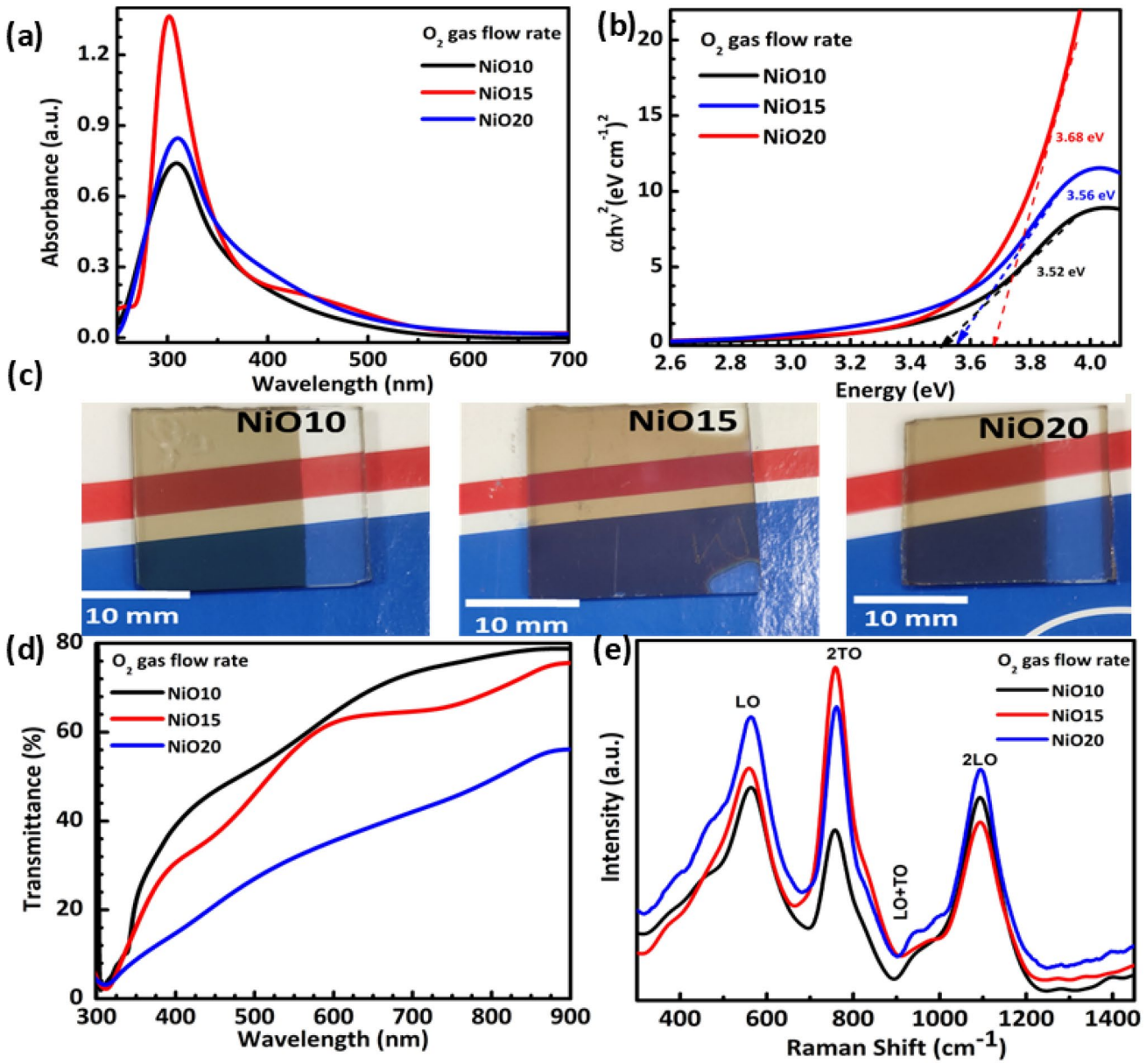


Fig.4 **a** Represents the absorption spectra of the films, while **b** represents the optical band gap measurements using a Tauc plot. **c** Optical images illustrate the transparency of the films, grown with the three flow rates of oxygen. **d** Transmittance spectra of the films, with a focus on the visible region of the spectrum. **e** Raman spectra of the NiO films

at 1094.26 cm^{-1}. The first and second-order modes represent the vibrations of the Ni–O bonds, and the sharp, intense band around 562.4 cm^{-1} shows the presence of defects in the form of Ni^{3+} ions along with the Ni^{2+} ions in the film, which affects the electrical characteristics of the developed photodetector.

3.3 Electrical characterization

The electrical characterization of the developed photodetector, based on the three different oxygen flow rates, was studied by both current-voltage (I-V) and transient current (I-t) measurements using a solar simulator with a light intensity of 1000 W/m^2. The I-V characteristics in the dark and illuminated conditions for the NiO films are summarised in Fig. 5. These were obtained over a narrow voltage region from -0.1 to 0.1 V, indicating the overall good electrical conductivity and the ability of the device to generate measurable currents even at low bias voltages. The NiO film was radiated from the top,

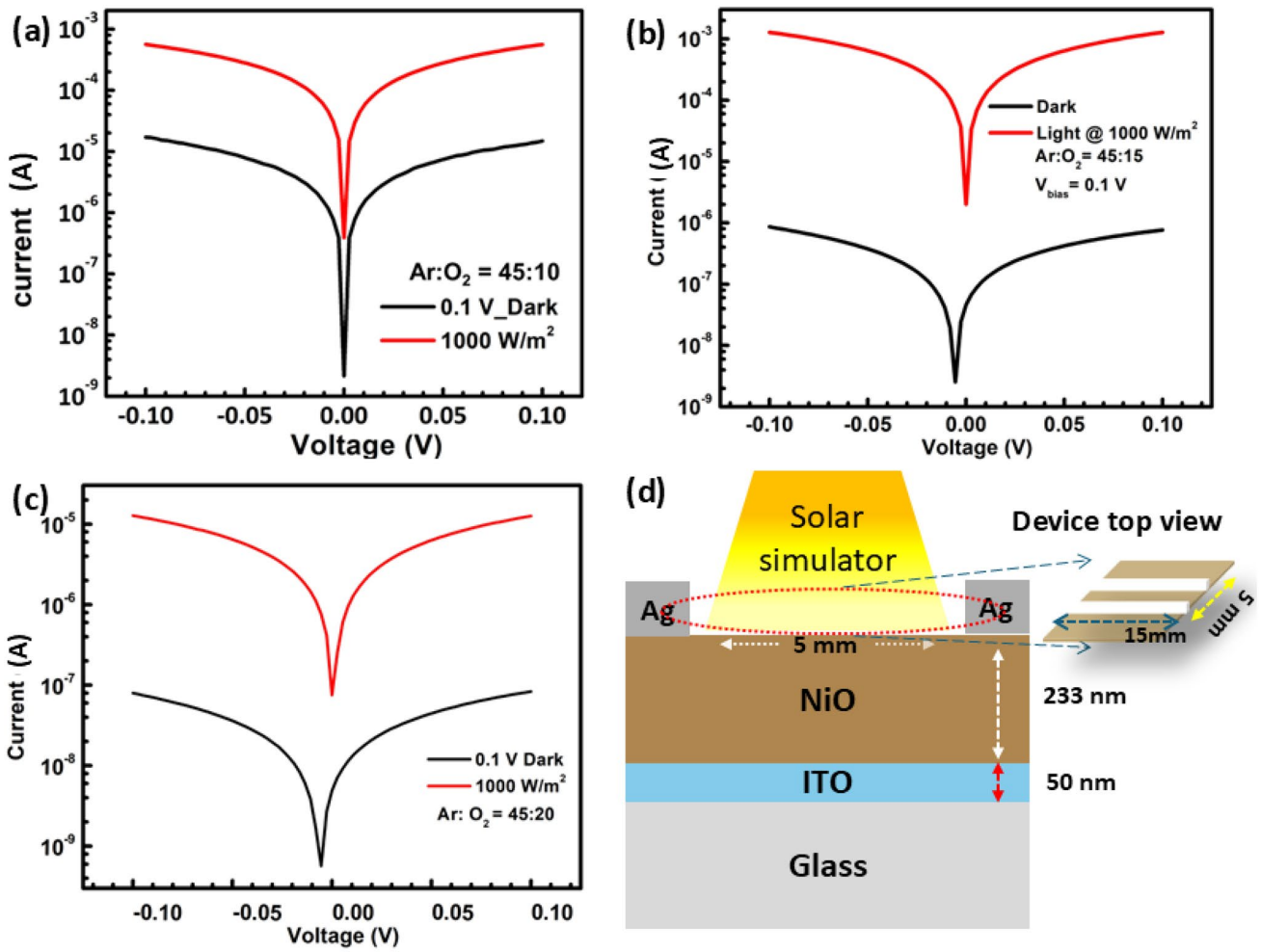


Fig. 5 I-V curves, under dark and light conditions, for the NiO film grown with **a** 10, **b** 15, and **c** 20 sccm O_2. The maximum enhancement was seen for the film grown in 15 sccm O_2. **d** Schematic of the device illuminated under light; all I-V analysis was done at 0.1 V. The voltage was applied between the two Ag electrodes deposited on the NiO films

as shown in the schematic in Fig. 5d. The maximum enhancement in the current (~1490 times) and correspondingly the highest sensitivity, at an applied voltage of 0.1 V, was for the film grown at an O_2 flow rate of 15 sccm.

The I-V analysis of the device was also carried out at different bias voltages, ranging from 0.1 to 5 V and the data is shown in Fig. 6. It was observed that once the bias voltage is increased, the dark current value increases, and hence the change in photocurrent was observed to be low. The ratios of photocurrent to dark current have been studied at different bias voltages, namely 0.1, 1, 2, and 5 and the corresponding values are 37.7, 13.1, 3.1, and 0.9 times for the NiO10 film. More importantly, it can be seen that at higher bias voltages the device I-V characteristics change from linear (ohmic) to rectifying behavior. The current enhancement was ~162 times for the film grown in 20 sccm O_2, while the enhancement was lowest at ~33 for 10 sccm O_2. As mentioned above, the maximum enhancement was obtained for the film grown at 15 sccm O_2. At lower oxygen concentrations, the defect densities are higher, which leads to a lower charge generation and collection. On the other hand, at the higher oxygen concentration, trap states are created in the material, which can also cause charge recombination.

The contact between ITO (*n*-type) and NiO (*p*-type) causes band bending at the interface [50]. This heterostructure formation also helps in charge separation, while the ohmic contact between the NiO and Ag allows for electrical conduction through the photodetector. Fig. 7 shows a schematic of the band bending at the heterojunction interface and charge transport mechanism between ITO, NiO, and Ag. The work function of silver is 4.26 eV [51], and NiO is about 5.3 eV [52]. At the metal-semiconductor (MS) interface, Fermi-level alignment takes place by charge transfer which results in band bending at the MS interface. While a Schottky-type contact is expected for this interface and seen at the higher bias voltages, at low bias voltages the device showed linear (ohmic) behavior. One possible reason for this could be the

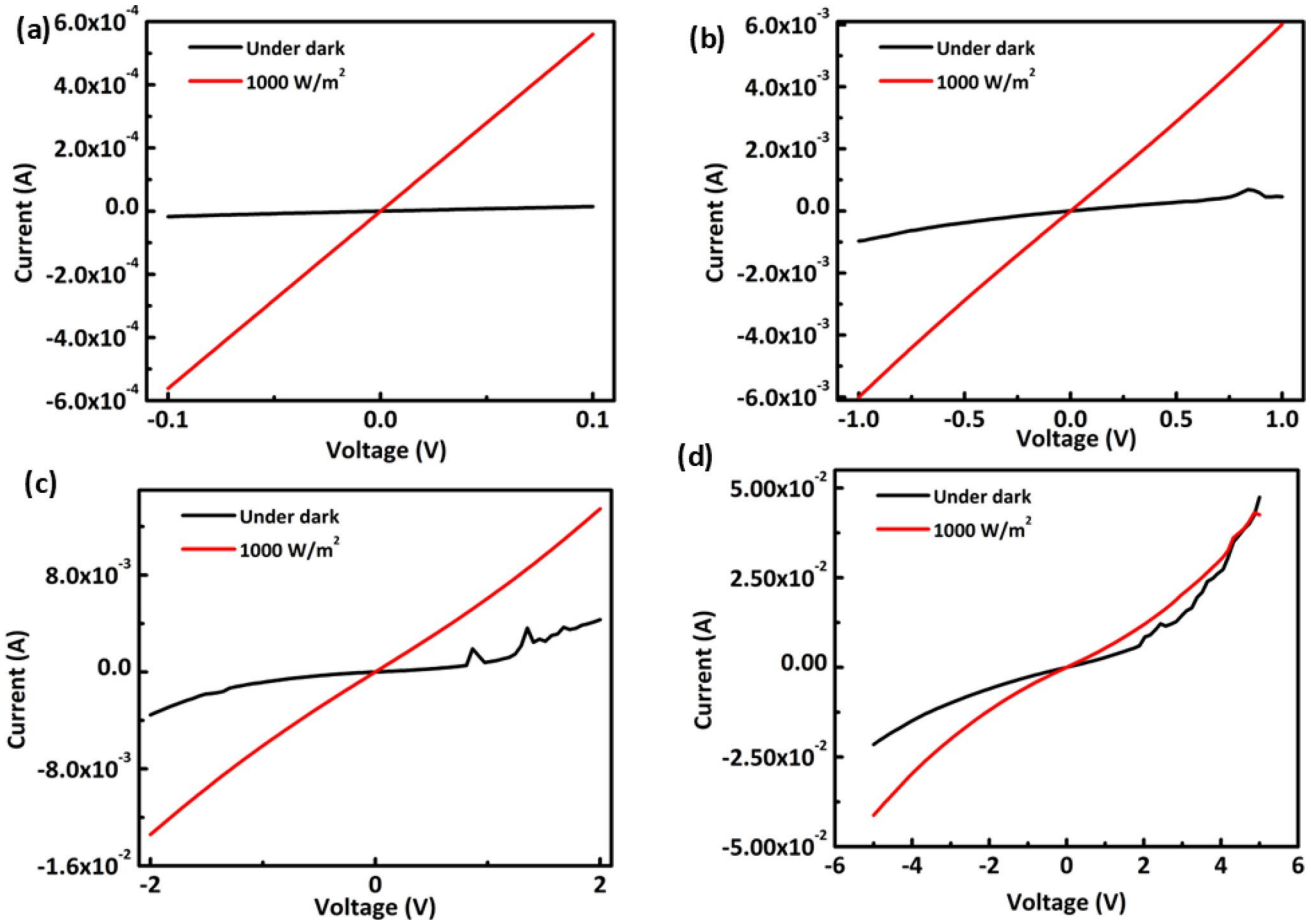


Fig. 6 Effect of bias voltage on the I-V curves, under dark and light conditions. The scans were recorded for the NiO10 film. The scan ranges are **a** 0.1 V, **b** 1 V, **c** 2 V, and **d** 5 V

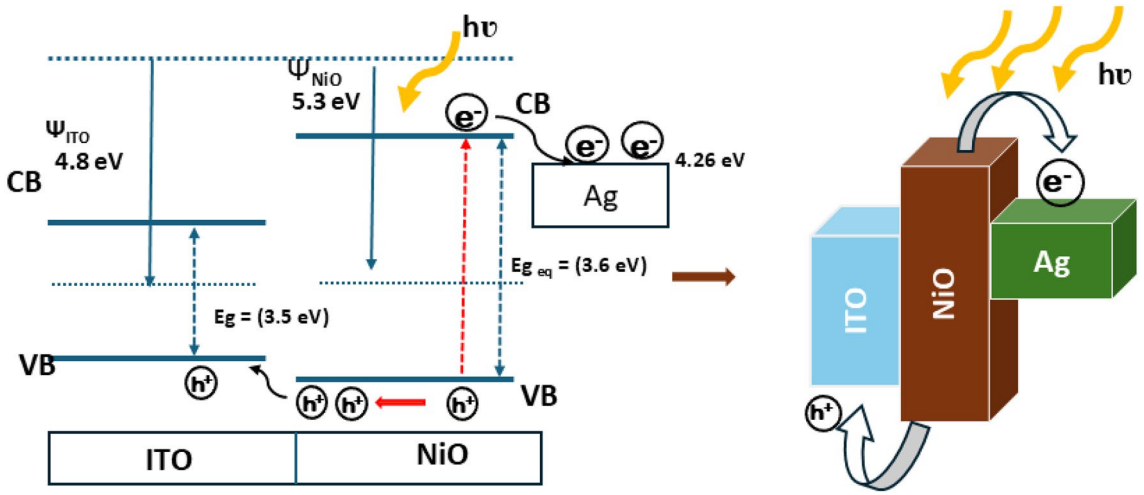


Fig. 7 Band bending at the MS interface supports charge transportation, and the contact between Ag and NiO facilitates Ohmic contact

Fermi level pinning at the NiO surface due to the defects created during deposition. This creates a thin MS interfacial layer through which electrons can tunnel creating this ohmic behavior, as shown in Fig.7. This type of behavior has been explored and well-understood in previous studies [53]. Further, to understand the role of the heterostructure, especially given the photoconductor device geometry we performed an I-V analysis of the device developed on a glass substrate

instead of ITO. The device geometry and the I-V behaviour are given in Fig. S3. Both NiO10 and NiO15, were investigated at bias voltages of 1, 5, and 10 V. There was a very low current observed under both dark and light conditions, which clearly shows that the heterostructure plays a significant role on charge separation and helps in better performance of the developed device.

Additionally, the photoresponse properties of the fabricated devices were examined under varying light intensities of the solar simulator. The various photocurrent responses from lower to higher illuminated light intensity have been shown in Fig 8. The photocurrent with respect to different illuminated powers demonstrates that the photocurrent linearly increased with the rise in light intensity from 550 to 1050 W/m^2. The NiO15 reveals a much better response with the increase in light intensity, compared to NiO10 and NiO20, indicating that this is the ideal film for photodetector applications.

3.4 Photodetector device performance metrics

The photodetector key figure of merit can be defined by responsivity R_λ that can be determined by following equation:

$$R_\lambda = \frac{\Delta I_\lambda}{P_\lambda A} \tag{3}$$

where ΔI_λ is the difference between photocurrent and dark current, P_λ is the incident light intensity, and A is the effective illuminated area. The spectral response was measured at different applied biases at 1000 W/m^2 light intensity. The responsivity of NiO10, NiO15, and NiO20 at different bias voltages are calculated and plotted in Fig. S4.

The devices' transient behavior was studied under pulse light illumination (width of 5 s) for the film grown with 10 and 15 sccm flow rate of oxygen while a longer pulse light illumination (width of 10 s) was used for the film grown with 20 sccm flow rate of oxygen. The choice for the pulse duration was necessitated due to the longer time needed for recombination for NiO20. The applied voltage used was 0.1 V and the study was performed under the same illumination source without variation of the other parameters. The transient response for the films are summarised in Fig. 9 and the difference can be seen in the nature of the curves for the different samples. Generally, response time for photoconductor geometry is usually slower and delayed and influenced by the active area between electrodes. The short pulse nature for NiO10 indicates that sufficient time for current saturation was not available, leading to a triangular shaped rise and decay current (Fig. 9a, b). The individual pulse was used to calculate the rise and decay times and the values obtained were 4.1 and 3.7 s respectively. This behaviour is related to the defects present in the film, primarily acting as trap centres for the charge carriers. For NiO15, the corresponding values are t_{rise}=0.5 s and t_{decay}= 1.4 s. The system also shows the expected rapid rise, saturation, and decay behaviour, though the slight decrease in current during saturation still indicates effect of some defects. For NiO20 the rise and fall times again increase, with values of t_{rise}=3.9 s and t_{decay}= 6.7 s. Among the three films, NiO15 exhibit the best response, in terms of the least rise and decay times. Coupled with the large responsivity and ideal I-V behaviour this film can be considered to the most optimised for heterojunction wide band gap photodetector behaviour. A comparison between device performance merits for photoconductor geometry rigid and flexible devices

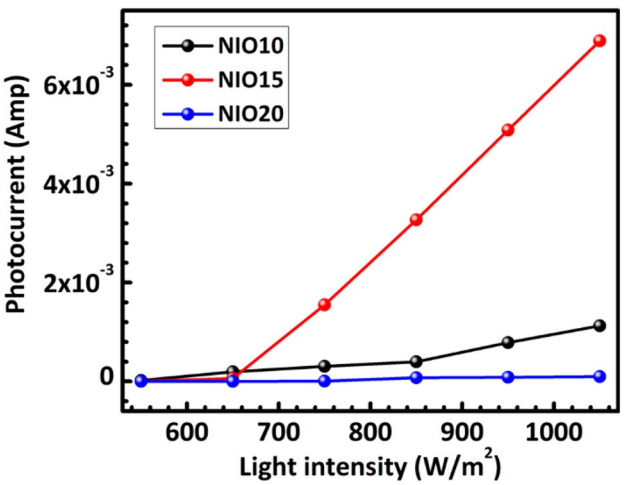


Fig. 8 The photocurrent versus the illuminated power intensity curves, at an applied bias 0.1 V, for NiO10, NiO15, and NiO20. The response of NiO15 is far superior, when compared to the other films

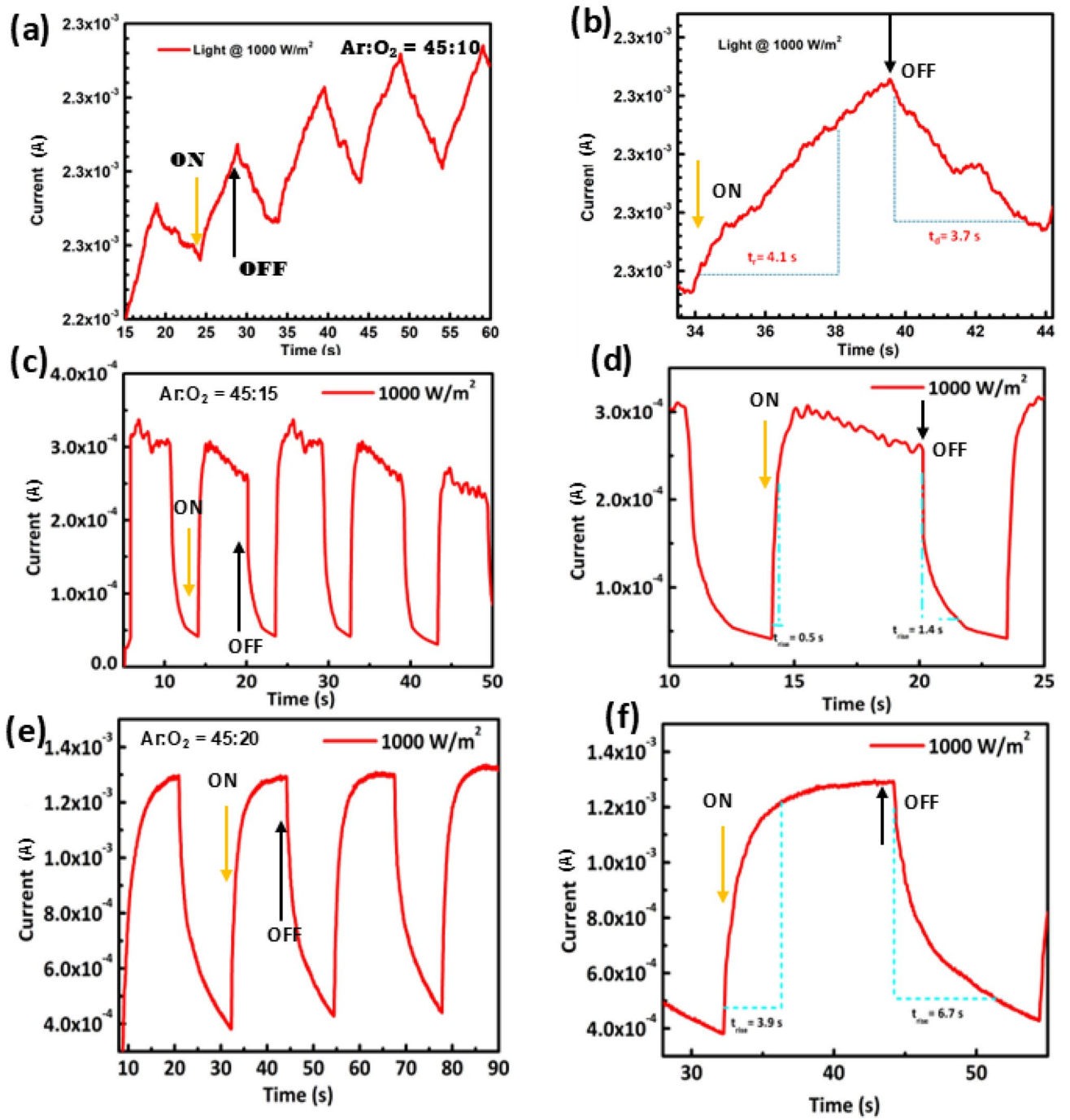


Fig. 9 The photo transient behaviour of the device under light and dark conditions, showing multiple and single cycle for **a**, **b** 10, **c**, **d** 15, and **e**–**f** 20 sccm O_2. The NiO film grown at 15 sccm O_2 has the shortest response time

reported in the literature with the present work is shown in Table 1. The fabricated device works well at a low bias voltage of 0.1 V, compared to other devices which require bias voltages of 1 V or more. The fabrication process is also simple with

Table 1 Comparison of the performance parameters of the fabricated detector with literature

Structure	Technique	Device	Bias	Response time	Wavelength	Responsivity	Reference
ITO/p-NiO/Ag (grown at 15 sccm O_2)	R-DC mag sputtering	Rigid	0.1 V	0.5 s/1.2 s	360 nm -Far IR	0.012 A/W	This work
Glass/n-ZnO/n-TiO_2/Al	Spin coat	Rigid	10 V	1.9 s/ 5 s	650 nm	0.57 A/W	[25]
Glass/n-SnO_2/n-TiO_2/Al	Spin coat	Rigid	10 V	3.7/1.3 s -0.1 s/2.1 s	650 nm	1.19 A/W	[23]
Ag/n-MoO/p-CuPc/CuPc: C6/ C60/BCP/Bphen/AGA	Thermal evaporation	Flexible	1 V	–	450–780 nm	–	[54, 59]
Si /Au/n-GaN/ITO	Laser MBE	Rigid	−3 V	ns	325 nm	0.27 A/W	[55, 60]
Si/p-BP/n-InSe/p-BP	EBL & exfoliation method	Rigid	1 V	3.5 µs/ 2.4 µs	405 nm & 520–1650 nm	27.8 A W	[56, 61]
n-ITO/p-SnS (QDs)	hydrothermal synthesis	Rigid	2 V	80 / 83 ms	390-780 nm	2.04 mA/W	[42, 48]
n-GaAs/Si	Epitaxially grown	Flexible	− 1 V	–	808 nm	~0.31 A/W	[57, 62]
n-MoS2/paper	Friction	Flexible	20 V	0.2–30 s	385–880 nm	1.5 µA /W	[58, 63]

only a single deposition step and contacts made ex-situ using silver epoxy. The response times are comparable or even better than the other devices, while the responsivity is also similar to other fabricated photodetectors.

4 Conclusion

In conclusion, a wide band gap photodetector was fabricated using reactively sputtered NiO, with different oxygen concentrations, keeping the argon flow rate constant. The developed photodetector with three O_2 flow rates exhibited good optoelectronic behavior in parallel electrode geometry. Among all three devices, NiO15-based device performs significantly better than the other two ratios, namely 10/45, and 20/45. Thus, this represents the optimised oxygen concentration for device performance. The work also shows that heterojunction formed with ITO has a significant effect on charge transportation and, hence, device performance. The NiO15 device has fast rise time of 0.5 s, with a slightly longer decay time of 1.4 s with responsivity 0.012 A/W. Given that NiO has a wide variety of other applications such as gas sensing and electrochromism, these films will also be tested for these applications as part of future work. Long term environmental stability and performance metrics of the device will also be evaluated. By modifying the sputtering parameters to eliminate the annealing step, the device can also be deposited on a flexible substrate. This, in turn, can be used to make wide band gap flexible photodetectors, which can be integrated into wearable devices.

Acknowledgements The work was supported by IIT Madras under the Institute of Eminence, India, Project Number SB/2223/1272/MM/328/ETWO/008702. The authors acknowledge the Department of Physics, IIT Madras, for XRD, FE-SEM, and Raman Spectroscopy measurement facilities, Industrial Consultancy and Sponsored Research (ICSR), IIT Madras for providing AFM facilities, and Department of Chemistry, IIT Madras for the UV-Visible spectroscopy measurements. We are also thankful to the Soft Matter and Complex Fluids Lab, MME, IIT Madras for the solar simulator facility.

Author contributions S.S. designed and performed the experiments, carried out the data analysis, and wrote the first draft of the manuscript. A.B. helped in the experimental work and supported data curation. P.S. supervised the overall work and reviewed and edited the manuscript.

Data availability Data sets generated during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

1. Alshanableh A, Albiss BA, Aljawrneh B, Alrousan S, Al-Othman A, Megdadi S. Novel and flexible asymmetric supercapacitors based on NiCo2O4 nanosheets coated on Al and Cu tapes for wearable devices applications. Disc Appl Sci. 2023;5:120. https://doi.org/10.1007/s42452-023-05341-8.
2. Bernardo CPCV, Lameirinhas RAM, Cunha JP, Torres S. A revision of the semiconductor theory from history to applications. Disc Appl Sci. 2024;6:316. https://doi.org/10.1007/s42452-024-06001-1.
3. Pirdaus NA, Ahmad N, Muhammad-Sukki F, Wan-Mohtar WAAQIJ. A narrative review on vital criteria of fungal dyes as dye-sensitized solar cell (DSSC). Disc Appl Sci. 2024;6(10):553. https://doi.org/10.1007/s42452-024-06256-8.
4. Hussain SZ, Ihrar M, Hussain SB, Oh WC, Ullah KJSAS. A review on graphene based transition metal oxide composites and its application towards supercapacitor electrodes. Disc Appl Sci. 2020;2:1–23. https://doi.org/10.1007/s42452-020-2515-8.
5. Chavali MS, Nikolova MPJS. Metal oxide nanoparticles and their applications in nanotechnology. SN Appl Sci. 2019;1:607. https://doi.org/10.1007/s42452-019-0592-3.
6. Chauhan P, Patel AB, Solanki G, Patel K, Pathak V, Sumesh C, Narayan S, Jha PKJ. Rhenium substitutional doping for enhanced photoresponse of n-SnSe2/p-Si heterojunction based tunable and high-performance visible-light photodetector. Disc Appl Sci. 2021;536:147739. https://doi.org/10.1007/s42452-021-03623-2.
7. Yu X, Marks TJ, Facchetti A. Metal oxides for optoelectronic applications. Nat Mat. 2016;15:383–96. https://doi.org/10.1038/nmat4599.

8. Razeghi M, Rogalski A. Semiconductor ultraviolet detectors. J Appl Phy. 1996;79:7433–73. https://doi.org/10.1063/1.362677.
9. Goldberg Y. Semiconductor near-ultraviolet photoelectronics. Semicond Sci Technol. 1999;14:R41. https://doi.org/10.1088/0268-1242/14/7/201.
10. Ohta H, Hosono H. Transparent oxide optoelectronics. Mater Today. 2004;7:42–51. https://doi.org/10.1016/S1369-7021(04)00288-3.
11. Monroy E, Omnès F, Calle F. technology, Wide-bandgap semiconductor ultraviolet photodetectors. Semicond Sci Techno. 2003;18:R33. https://doi.org/10.1088/0268-1242/18/4/201.
12. Ahmed AA, Devarajan M, Afzal N, Physical AA. Fabrication and characterization of high performance MSM UV photodetector based on NiO film. Actuators A Phys. 2017;262:78–86. https://doi.org/10.1016/j.sna.2017.05.028.
13. Alaie Z, Nejad SM, Yousefi M. Recent advances in ultraviolet photodetectors. Mater Sci Semicond Process. 2015;29:16–55. https://doi.org/10.1016/j.mssp.2014.02.054.
14. Patel A, Pataniya P, Narayan S, Sumesh C, Pathak V, Solanki G, Patel K, Jha PK. Investigation of structural, electrical and optical properties of SbXW1-XSe2 single crystals. Mater Sci Semicond Process. 2018;81:108–12. https://doi.org/10.1016/j.mssp.2018.01.012.
15. Patel M, Park H-H, Bhatnagar P, Kumar N, Lee J, Kim JJ. Transparent integrated pyroelectric-photovoltaic structure for photo-thermo hybrid power generation. Nat Commun. 2024;15:3466. https://doi.org/10.1038/s41467-024-47483-2.
16. Kong X, Liu C, Dong W, Zhang X, Tao C, Shen L, Zhou J, Fei Y, Ruan S. Metal-semiconductor-metal TiO2 ultraviolet detectors with Ni electrodes. Appl Phys Lett. 2009. https://doi.org/10.1063/1.3103288.
17. Bai Z, Yan X, Chen X, Zhao K, Lin P, Zhang Y. High sensitivity, fast speed and self-powered ultraviolet photodetectors based on ZnO micro/nanowire networks. Prog Nat Sci Mater Int. 2014;24:1–5. https://doi.org/10.1016/j.pnsc.2014.01.01.
18. Inamdar S, Rajpure K. compounds, High-performance metal–semiconductor–metal UV photodetector based on spray deposited ZnO thin films. J Alloys Compd. 2014;595:55–9. https://doi.org/10.1016/j.jallcom.2014.01.147.
19. Fang X, Hu L, Huo K, Gao B, Zhao L, Liao M, Chu PK, Bando Y, Golberg D. New ultraviolet photodetector based on individual Nb2O5 nanobelts. Adv Funct Mater. 2011;21:3907–15. https://doi.org/10.1002/adfm.201100743.
20. Müller A, Konstantinidis G, Dragoman M, Neculoiu D, Dinescu A, Androulidaki M, Kayambaki SAM, Vasilache D, Buiculescu C. GaN membrane-supported UV photodetectors manufactured using nanolithographic processes. Microelectron J. 2009;40:319–21. https://doi.org/10.1016/j.mejo.2008.07.021.
21. Yang W, Zhang F, Liu Z, Wu Z. Effects of annealing on the performance of 4H-SiC metal–semiconductor–metal ultraviolet photodetectors. Mater Sci Semicond Process. 2008;11:59–62. https://doi.org/10.1016/j.mssp.2008.11.001.
22. Pramanik P, Sen S, Singha C, Roy AS, Das A, Sen S, Bhattacharyya A. Compositional inhomogeneities in AlGaN thin films grown by molecular beam epitaxy: effect on MSM UV photodetectors. J Appl Phys. 2016. https://doi.org/10.1063/1.4964420.
23. Pandey U, Pal N, Ghosh A, Suman S, Biring S, Pal BN. Blue sensitive sub-band gap negative photoconductance in SnO 2/TiO 2 NP bilayer oxide transistor. Nanoscale. 2024;16:8504–13. https://doi.org/10.1039/D4NR00406J.
24. Saroha J, Semalti P, Tanwar P, Kumar M, Sharma SN. Investigating bifunctional linker-assisted photocatalytic behavior of Ag–TiO2 nanocomposites. J Appl Phys. 2024;136:5. https://doi.org/10.1063/5.0213838.
25. Suman S, Dahiya S, Jaiswal RP, Swaminathan P, Pal BN. Fabrication of a red-sensitive heterojunction photodetector by using a narrowband organic dye. J Phys Chem C. 2023;127:19182–8. https://doi.org/10.1021/acs.jpcc.3c04112.
26. Suman S, Pandey U, Pal N, Swaminathan Pal BN. narrowband red-sensitive positive—negative reversible photodetector using organic dye on the TiO2/SnO2 heterojunction. ACS Photonics. 2023;11:93–101. https://doi.org/10.1021/acsphotonics.3c01009.
27. Suman S, Swaminathan P, Pal BN, Organic dye based longer wavelength photodetector for narrowband application, 2022 IEEE International Conference on Emerging Electronics (ICEE), IEEE, 2022, pp. 1–4.
28. Li Z, Li Z, Zuo C, Fang X. Application of nanostructured TiO2 in UV photodetectors: a review. Adv Mat. 2022;34:2109083. https://doi.org/10.1002/adma.202109083.
29. Krasienapibal TS, Fukumura T, Hirose Y, Hasegawa T. Improved room temperature electron mobility in self-buffered anatase TiO2 epitaxial thin film grown at low temperature. Jpn J Appl Phys. 2014;53:9.
30. Zhang J, Hoye Li W, MacManus-Driscoll JR, Budde M, Bierwagen O, Wang L, Du Y, Wahila M, Piper L. Electronic and transport properties of Li-doped NiO epitaxial thin films. J Mater Chem C. 2018;6:2275–82. https://doi.org/10.1039/C7TC05331B.
31. Parui SS, Kumar N, Tiwari P, Tiwari N, Chauhan R. Zinc oxide and cupric oxide based thin films for solar cell applications. Mater Today Proc. 2021;41:233–6. https://doi.org/10.1016/j.matpr.2020.08.027.
32. Di Girolamo D, Di Giacomo F, Matteocci F, Marrani AG, Dini D, Abate A. Progress, highlights and perspectives on NiO in perovskite photovoltaics. Chem Sci. 2020;11:7746–59. https://doi.org/10.1039/d0sc02859.
33. Lu H, Yuan X, Chen B, Gong C, Zeng H, Wei X. Technology, Self-rectifying resistive switching device based on n-ZnO/p-NiO junction. J Sol-Gel Sci Technol. 2017;82:627–34. https://doi.org/10.1007/s10971-017-4344-9.
34. Ahn Y, Son JY. Resistive random access memory characteristics of NiO thin films with an oxygen-deficient NiO 095 layer. J Korean Ceram Soc. 2021;47:9342–6.
35. Karsthof R, Anton AM, Kremer F, Grundmann M. Nickel vacancy acceptor in nickel oxide: doping beyond thermodynamic equilibrium. Phys Rev Materials. 2020;4: 034601. https://doi.org/10.1103/PhysRevMaterials.4.034601.
36. Salunkhe P, Kekuda D. Structural, spectroscopic and electrical properties of dc magnetron sputtered NiO thin films and an insight into different defect states. Phys A. 2021;127:390. https://doi.org/10.1007/s00339-021-04501-0.
37. Wang M. Thimont Y, Presmanes L, Diao X, Barnabé A, The effect of the oxygen ratio control of DC reactive magnetron sputtering on as-deposited non stoichiometric NiO thin films. Appl Surf Sci. 2017;419:795–801. https://doi.org/10.1016/j.apsusc.2017.05.095.
38. Lahiji FAF, Bairagi S, Magnusson R, Sortica MA, Primetzhofer D, Ekström E, Paul B, Le FA, Eklund P. Growth and optical properties of NiO thin films deposited by pulsed dc reactive magnetron sputtering. J Vac Sci Technol A. 2023;41:6. https://doi.org/10.1116/6.0002914.
39. Sun H, Jen S-U, Chen S-C, Ye S-S, Wang X. The electrical stability of In-doped ZnO thin films deposited by RF sputtering. J Phys D Appl Phys. 2016;50: 045102. https://doi.org/10.1088/1361-6463/aa5002.
40. Wang L, Yang Y, Jin S, Marks T. MgO (100) template layer for CdO thin film growth: Strategies to enhance microstructural crystallinity and charge carrier mobility. J Appl Phys. 2006. https://doi.org/10.1063/1.2195093.

41. Sun H, Chen S-C, Hsu S-W, Wen C-K, Chuang T-H, Wang X. Microstructures and optoelectronic properties of nickel oxide films deposited by reactive magnetron sputtering at various working pressures of pure oxygen environment. Ceram Int. 2017;43:S369–75. https://doi.org/10.1016/j.ceramint.2017.05.242.
42. Kim H, Kim K, Hong J. Effect of oxygen-evaporation-preventative post-annealing gas conditions on NiO thin film. Coatings. 2023;11:1954. https://doi.org/10.3390/coatings13111954.
43. Jadoon NAK, Puvanenthiram V, Mosa MAH, Sharma A, Wang K. Recent advances in aluminum nitride (AlN) growth by magnetron sputtering techniques and its applications. Inorganics. 2024;12:264. https://doi.org/10.3390/inorganics12100264.
44. Schnitter C, Rosen HH. Epitaxial growth of magnetron-sputtered ZrB2 films on Si (100) substrates. Phys Status Solidi A. 2022;219:2200330. https://doi.org/10.1002/pssa.202200330.
45. Fang Y-S, Do TH, Chiu K-A, Chen W-C, Chang L. Reactive sputtering deposition of epitaxial TiC film on Si (100) substrate. Coatings. 2020;10:647. https://doi.org/10.3390/coatings10070647.
46. Tao J, Chai J, Lu X, Wong LM, Wong TI, Pan J, Xiong Q, Chi D, Wang S. Growth of wafer-scale MoS 2 monolayer by magnetron sputtering. Nanoscale. 2015;7:2497–503. https://doi.org/10.1039/C4NR06411A.
47. Singh DK, Pant R, Chowdhury AM, Roul B, Nanda KK, Krupanidhi SB. Defect-mediated transport in self-powered, broadband, and ultrafast photoresponse of a MoS2/AlN/Si-based photodetector. ACS Appl Electron Mater. 2020. https://doi.org/10.1021/acsaelm.0c00007.
48. Modi KH, Pataniya PM, Patel V, Sumesh C. Self-powered photodetector functionalized by SnS quantum dots. Opt Mater. 2022;129: 112504. https://doi.org/10.1016/j.optmat.2022.112504.
49. Patel A, Pataniya P, Solanki G, Sumesh C, Patel K, Pathak V. Microstructures, Fabrication, photoresponse and temperature dependence of n-VO2/n-MoSe2 heterojunction diode. Superlatt Microstruct. 2019;130:160–7. https://doi.org/10.1016/j.spmi.2019.04.032.
50. Rana AK, Kumar M, Ban DK, Wong CP. Yi J, Kim, J, Enhancement in performance of transparent p-NiO/n-ZnO heterojunction ultrafast self-powered photodetector via pyro-phototronic effect. Adv Electron Mater. 2019;5:1900438. https://doi.org/10.1002/aelm.201900438.
51. Sekertekin B, Ozel K, Atilgan A, Yildiz A. Ultraviolet photodiode fabricated from TiO2 nanorods/p-silicon heterojunction. Mater Lett. 2022;323: 132565. https://doi.org/10.1016/j.matlet.2019.10.109.
52. Patel M, Pataniya PM, Sumesh C. Enhanced photoresponse by plasmon resonance in Ni-WS2/Si photodiode. Mater Res Bull. 2022;145: 111518. https://doi.org/10.1016/j.materresbull.2021.111518.
53. Ahmed AA. Hashim M, Abdalrheem R, Rashid M, Compounds, High-performance multicolor metal-semiconductor-metal Si photodetector enhanced by nanostructured NiO thin film. J Alloys Compd. 2019;798:300–10. https://doi.org/10.1016/j.jallcom.2019.05.286.
54. Terlemezoglu M, Surucu O, Isik M, Gasanly NM, Parlak M. Temperature-dependent optical characteristics of sputtered NiO thin films. Appl Phys A. 2022;128:50. https://doi.org/10.1007/s00339-021-05197-y.
55. Alzahrani AOM, Abdel-Wahab M, Alayash M, Aida M. Metals and ITO contact nature on ZnO and NiO thin films. Braz J Phys. 2021;51:1159–65. https://doi.org/10.1007/s13538-021-00905-8.
56. Drummond TJ. Work functions of the transition metals and metal silicides, Sandia National Lab(SNL-NM), Albuquerque, NM (United States); Sandia. J Appl Phys. 1999.
57. Yip H-L, Jen AK-YE. Science, recent advances in solution-processed interfacial materials for efficient and stable polymer solar cells. Energy Environ Sci. 2012;5:5994–6011. https://doi.org/10.1039/C2EE02806A.
58. Abdy H, Aletayeb A, Kolahdouz M, Soleimani EAJAA. Investigation of metal-nickel oxide contacts used for perovskite solar cell. AIP Adv. 2019. https://doi.org/10.1063/1.5063475.
59. Wang Y, Li X, Liu S, Zhang L, Xie W. Flexible top-illuminated organic photodetector using an ultrathin-metal-based transparent electrode. Photon Res AIP Adv. 2023;11:2100–4. https://doi.org/10.1364/PRJ.502731.
60. Yadav G, Gupta V. Tomar M, Double Schottky metal–semiconductor–metal based GaN photodetectors with improved response using laser MBE technique.J. Mater Res. 2022;37:457–69. https://doi.org/10.1557/s43578-021-00467-0.
61. Song Z, Wei B, Wang Q, Wang W, Cao Z, Zhang L, Mu Q, Han T, Li F, Zhu X. High-performance metal electrode-enhanced double parallel p–n heterojunctions photodetector. J Appl Phys. 2023;133:12. https://doi.org/10.1063/5.0141523.
62. Hong ChuRJ, Kang SS, Ryu G, Han J-H, Yu KJ, Jung D, Choi W. Flexible GaAs photodetector arrays hetero-epitaxially grown on GaP/Si for a low-cost III-V wearable photonics platform. Opt Express. 2020;28:36559–67. https://doi.org/10.1063/5.0141523.
63. Mazaheri A, Lee M, Van Der Zant HS, Frisenda R, Castellanos-Gomez A. MoS 2-on-paper optoelectronics: drawing photodetectors with van der Waals semiconductors beyond graphite. Nanoscale. 2020;12:19068–74. https://doi.org/10.1039/D0NR02268C.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.